

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question

Body Length, Filter Time, and Lunges per Dive for Four Whale Species

Whale species	Typical adult body length (meters)	Average time to filter all engulfed water (seconds)	Average number of lunges per dive deeper than 50 meters
fin	18–22	31.30	3.95
humpback	11–17	17.12	6.28
minke	7–10	8.88	7.48
blue	24–34	60.27	4.02

Some whale species practice lunge feeding, in which they lunge toward prey with their mouths open at wide angles, collect the prey and the surrounding water, and then filter out the water through baleen plates in their mouths. Although the volume of water engulfed increases with whales’ body length, the surface area of whales’ baleen plates, which influences the rate at which water can be filtered, does not increase with body length to the same degree, which helps explain why _____

Which choice most effectively uses data from the table to complete the statement?

Answer

- A. minke whales and humpback whales show similar average filter times.
- B. humpback whales show an average of 6.28 lunges per dive.
- C. fin whales show a longer average filter time than minke whales do.
- D. blue whales show the longest average filter time and the highest average number of lunges per dive.

Correct Answer: C

Rationale

Choice C is the best answer. To support the claim, we need to show that longer whales take more time to filter all the water they engulf than shorter whales do. This choice accurately reflects that a longer whale (the fin whale) takes more time to filter engulfed water (31.30 seconds on average) compared to a shorter whale (the minke whale, which only took 8.88 seconds on average).

Choice A is incorrect. The table shows that minke whales take an average of 8.88 seconds to filter engulfed water, while humpback whales take an average of 17.12 seconds to complete the same task. Choice B is incorrect. This choice doesn’t reflect the claim about baleen plates. The claim explains why whales of differing lengths take different amounts of time to filter engulfed water. This choice doesn’t compare whales of different lengths, and it focuses on the number of lunges, which isn’t shown to be relevant to filter time. Choice D is incorrect. The table shows that blue whales average 4.02 lunges per dive, which is not the highest average among the whales in the table.